CENG381 Stochastic Processes

Martingales and Optional Stopping Theorem - Answer Key 1

Q1 Solution

a) $S(n)$ is a martingale:

$$
\begin{aligned}
E[S(n+1) \mid F_n] &= E[S(n) + X_{n+1} \mid F_n] \\
&= E[S(n) \mid F_n] + E[X_{n+1} \mid F_n] \\
&= S(n) + E[X_{n+1}] \quad (S(n) \text{ is } F_n\text{-measurable;} \\
&\qquad\qquad X_{n+1} \text{ is independent of } F_n) \\
&= S(n) + 0 \\
&= S(n) \quad \text{(OK)}
\end{aligned}
$$

So $S(n)$ satisfies $E[S(n+1)|F_n] = S(n)$: it is a martingale.

b) $M(n) = S(n)^2 - n$ is a martingale:

Compute $E[M(n+1) \mid F_n]$:

$$
\begin{aligned}
& E[S(n+1)^2 - (n+1) \mid F_n] \\
&= E[(S(n) + X_{n+1})^2 \mid F_n] - (n+1) \\
&= E[S(n)^2 + 2 \cdot S(n) \cdot X_{n+1} + X_{n+1}^2 \mid F_n] - (n+1) \\
&= S(n)^2 + 2 \cdot S(n) \cdot E[X_{n+1}] + E[X_{n+1}^2] - (n+1) \\
&\quad [\text{using } S(n) \text{ is } F_n\text{-measurable, } X_{n+1} \text{ independent of } F_n] \\
&= S(n)^2 + 2 \cdot S(n) \cdot 0 + 1 - (n+1) \\
&= S(n)^2 - n \\
&= M(n) \quad \text{(OK)}
\end{aligned}
$$

So $M(n) = S(n)^2 - n$ is a martingale.

c) Classification when $E[X_i] = c > 0$:

$E[S(n+1) \mid F_n] = S(n) + E[X_{n+1}] = S(n) + c > S(n)$.

Since expected future value EXCEEDS current value, $S(n)$ is a SUBMARTINGALE.
(The walk has upward drift -- on average it grows.)

For $-S(n)$: $E[-S(n+1) \mid F_n] = -S(n) - c < -S(n)$.
So $-S(n)$ is a SUPERMARTINGALE.
(Flipping the sign reverses the inequality.)

Q2 Solution

a) Winning probability h(k) = P(reach N before 0):

Apply OST to the martingale S(n). Since T is bounded (T <= N^2 can be shown), the conditions are satisfied:

$$E[S(T)] = E[S(0)] = k$$

At stopping time T, S(T) = N (with probability h(k)) or S(T) = 0 (with probability 1 - h(k)). So:

$$E[S(T)] = N * h(k) + 0 * (1 - h(k)) = N * h(k)$$

Setting equal:

$$N * h(k) = k$$
$$h(k) = k / N$$

The probability of reaching N before 0 is exactly k/N -- proportional to the starting position. This is the classic Gambler's Ruin result for a symmetric random walk.

b) Expected absorption time E[T | S(0) = k]:

Apply OST to the martingale M(n) = S(n)^2 - n:

$$E[M(T)] = E[M(0)] = k^2 - 0 = k^2$$

At time T:

$$M(T) = S(T)^2 - T$$

$$\begin{aligned} E[M(T)] &= E[S(T)^2] - E[T] \\ &= [N^2 * h(k) + 0^2 * (1-h(k))] - E[T] \\ &= N^2 * (k/N) - E[T] \\ &= N*k - E[T] \end{aligned}$$

Setting equal to k^2:

$$N*k - E[T] = k^2$$

E[T] = N*k - k^2 = k*(N - k)

c) Numerical answer for N=4, k=1:

h(1) = 1/N = 1/4 = 0.25

There is only a 25% chance of reaching the goal (N=4) before going broke, starting just one step away from ruin.

E[T] = k*(N-k) = 1*(4-1) = 3 steps

The game lasts an average of 3 steps. This seems short, but makes sense: 75% of the time the gambler goes broke quickly (loses in 1 step with prob 1/2, or in a few steps), and only 25% of the time does the game last long enough to reach 4.

Q3 Solution

a) T* is a valid stopping time:

T* = min{n >= 1 : S(n) > 0}.

The event {T* = n} means:
S(1) <= 0, S(2) <= 0, ..., S(n-1) <= 0, S(n) > 0.

Each of these conditions depends only on X_1, ..., X_n (not on future steps), so {T* = n} is determined by F_n. Therefore T* is a valid stopping time: the decision to stop is based only on past outcomes.

b) Value of S at stopping:

Each step X_i is either +1 or -1. So S(n) always changes by exactly 1 at each step. The first time S(n) becomes positive (> 0), starting from S(0) = 0, the walk must reach +1 (it cannot jump from 0 or below to +2 or higher in a single step of size +1).

Therefore: S(T*) = +1 whenever T* < inf.

The gambler always stops exactly 1 unit ahead when they stop.

c) Which OST condition fails? E[T*] = infinity:

We have E[S(T*)] = 1 (since S(T*)=1 always when finite), but E[S(0)]=0.
This contradicts OST unless one of its conditions fails.

Condition (ii) requires E[T*] < inf with bounded increments.
Increments are bounded (|X_i| = 1), so the issue must be E[T*] = inf.

Why E[T*] = inf for symmetric random walk:

The symmetric random walk on Z is RECURRENT: it visits every integer infinitely often with probability 1. In particular, it returns to 0 infinitely often.

The expected return time to state 0 for a symmetric random walk is INFINITE (the walk is null-recurrent). Since T* is related to the return time, E[T*] = inf.

Conclusion: OST does NOT apply here because E[T*] = inf.
The 'Stop When You're Ahead' strategy beats the fair game in expectation of the final amount -- but only by waiting an infinitely long time on average. No practical advantage exists.